Spectral Pauli Pruning: Efficient Feature Selection for Flipped Quantum Hamiltonian Classifiers

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***Abstract*—Quantum Machine Learning (QML) promises computational advantages but faces significant bottlenecks in data encoding, measurement complexity, and training trainability (barren plateaus). Recent *flipped models*, such as the Hamiltonian Classifier, address the encoding bottleneck by representing data as quantum observables (Hamiltonians) rather than states. However, the *Simplified Hamiltonian (SIM)* variant relies on a simplified decomposition of the Hamiltonian into Pauli strings, where the selection of these strings is typically random or heuristic, potentially requiring large sets for adequate accuracy. In this work, we address the research question: *What is the smallest set of Pauli strings that generates adequate accuracy?* We propose a novel Spectral Moment Pauli Generator (Algorithm 3) to identifying the spectrally optimal Pauli basis. By decomposing the class-conditional covariance difference matrix into the Pauli basis, we identify the interaction terms that maximize class separation energy. We validate this approach on the 20 Newsgroups dataset, achieving 83.49% accuracy with only 57.6% of the full Pauli basis (adaptive $\eta = 0.98$), outperforming the full-basis model. We further demonstrate that a massive 94% compression (keeping only top-256 of 4096 terms) retains 99% of the optimal accuracy. Our results confirm that noise in the Hilbert space resides in the spectral tails, and targeted pruning allows for highly efficient, scalable quantum classifiers.**

***Index Terms*—Quantum Machine Learning, Hamiltonian Classifier, Flipped Models, Pauli Decomposition, Feature Selection, NISQ**

I. Introduction

Quantum computing offers the potential for exponential speedups in specific computational tasks such as integer factorization [12] and Hamiltonian simulation [11]. In the domain of Machine Learning, Variational Quantum Circuits (VQCs) have emerged as the leading paradigm for Noisy Intermediate-Scale Quantum (NISQ) devices [3], [4].

However, VQCs face three critical bottlenecks:

1) **Data Encoding**: Loading classical data $x \in \mathbb{R}^d$ into a quantum state $|\psi(x)\rangle$ requires either exponential gate depth (amplitude encoding) or linear qubit scaling (angle encoding) [5], [10].
2) **Measurement Complexity**: Extracting information requires projective measurements. Estimating an expectation value to precision ϵ requires $O(1/\epsilon^2)$ shots, which becomes prohibitive for complex operators [6].
3) **Barren Plateaus**: The loss landscapes of deep parameterized quantum circuits often exhibit vanishing gradients, making training exponentially hard [7].

To mitigate these issues, *flipped models* [2] have been proposed. In a flipped model, classical data is not encoded into the state but into the observable (Hamiltonian) $H(x)$. This effectively "classifies the data" by measuring it against a learned quantum state $|\psi(\theta)\rangle$. Tiblias et al. [1] introduced the *Hamiltonian Classifier (HAM)* and its variants, including the *Simplified Hamiltonian (SIM)*. While SIM achieves logarithmic qubit complexity, the set of Pauli strings $\{P_j\}$—which defines the feature space—is typically chosen randomly.

In this paper, we focus on the **Feature Selection problem for Quantum Observables**. We ask: *Can we rigorously derive the minimal set of Pauli strings required to capture the classification boundaries of a dataset?*

We introduce **Spectral Pauli Pruning**, a method based on the eigendecomposition of the data covariance matrix. We demonstrate that the relevant information is sparse in the Pauli basis, allowing for massive compression without accuracy loss.

II. Related Work

A. Quantum Kernels and Feature Maps

Schuld et al. [9] established that VQCs are equivalent to kernel methods in an exponentially large feature space. Havlicek et al. [8] proposed quantum-enhanced feature spaces using ZZ interactions. Our work differs by optimizing the *measurement observable* rather than the feature map circuit, allowing classical preprocessing to determine the optimal quantum interactions.

B. Classical Shadows

Huang et al. [6] introduced Classical Shadows to efficiently estimate properties of quantum states with few measurements. Jerbi et al. [2] combined shadows with flipped models to avoid full tomography. Our Spectral Pruning complements this by reducing the number of observables that *need* to be estimated in the first place, further reducing the sample complexity.

C. Hamiltonian Learning

In quantum many-body physics, learning the Hamiltonian from dynamics is a well-studied problem [13]. Here, we invert the problem: we construct a Hamiltonian that best separates classical data classes. The theoretical foundation is similar to finding an optimal witness operator for state discrimination.

III. Methodology

A. Flipped Models and SIM Formulation

In a standard flipped model, the probability of class 1 is given by:

$$f(x) = \sigma\left(\mathrm{Tr}(\rho_\theta H(x))\right) \tag{1}$$

where $\rho_\theta = |\psi_\theta\rangle\langle\psi_\theta|$ is the parametrized quantum state and $H(x)$ is the data-encoding Hamiltonian. For the SIM variant, we define the Hamiltonian as:

$$H(x) = \sum_{j=1}^{p} (x^T P_j x) w_j P_j \tag{2}$$

where x is the processed input vector (e.g. padded and normalized), P_j are n-qubit Pauli strings (elements of $\{I, X, Y, Z\}^{\otimes n}$), and w_j are learnable classical weights. The expectation value becomes:

$$\langle H(x)\rangle_{\psi_\theta} = \sum_{j=1}^{p} \underbrace{(x^T P_j x)}_{\text{Classical Projection}} \cdot w_j \cdot \underbrace{\langle\psi_\theta|P_j|\psi_\theta\rangle}_{\text{Quantum Bias}} \tag{3}$$

This reveals that the model effectively learns a linear combination of classical quadratic features $x^T P_j x$, modulated by both a classical weight w_j and a quantum bias $\langle P_j\rangle_\theta$.

B. Proposed Algorithm: Spectral Moment Pauli Generator

The dimensionality of the Pauli group $\mathcal{P}_n$ is 4^n. We propose that the most informative Pauli observables for classification are those that align with the directions of maximum variance difference between classes. We define the *Class-Conditional Covariance Difference* matrix Δ:

$$\Delta = \text{Cov}(X|y=1) - \text{Cov}(X|y=0) \tag{4}$$

This matrix captures the quadratic energy difference between the two classes. If Δ is positive definite along the direction of an operator P, then measuring P will yield statistically distinguishable values for the two classes.

Since Pauli matrices form an orthogonal basis for Hermitian matrices (under the Hilbert-Schmidt inner product $\langle A, B\rangle = \frac{1}{2^n}\text{Tr}(A^\dagger B)$), we can decompose Δ uniquely:

$$\Delta = \sum_{P\in\mathcal{P}_n} c_P P \tag{5}$$

where the spectral coefficient c_P is:

$$c_P = \frac{1}{2^n}\text{Tr}(\Delta P) \tag{6}$$

The magnitude $|c_P|$ directly quantities the *Information Gain* provided by measuring the Pauli word P.

1) Algorithm Steps:

1) Compute empirical covariances $\hat{\Sigma}_0, \hat{\Sigma}_1$ for classes 0 and 1.
2) Compute $\Delta = \hat{\Sigma}_1 - \hat{\Sigma}_0$.
3) For all (or a subset of) Pauli strings P, compute coefficients c_P.
4) Sort strings by $|c_P|$ descending.
5) **Adaptive Cutoff**: Select smallest k such that the cumulative energy ratio exceeds η:

$$\frac{\sum_{i=1}^{k} |c_{(i)}|}{\sum_{j=1}^{4^n} |c_j|} \geq \eta \tag{7}$$

IV. Exploratory Analysis and Hypothesis Generation

Before defining our final spectral algorithm, we conducted a series of preliminary experiments to understand the distribution of information within the Pauli group.

A. Phase 1: Ablation and Transferability

We first investigated whether a "universal" set of Pauli strings exists. We trained a full-basis Exact SIM model on a biological dataset (E. Coli) and ranked the Pauli strings by their effective weight contribution $|w_j\langle P_j\rangle|$. We extracted the **Top-32** performing strings and evaluated their performance on disparate datasets (Wine, MNIST). *Result:* While some low-order strings (e.g., single qubit Z_i) showed robustness, the specific high-order correlations were dataset-dependent. This suggested that a static subset is insufficient for general tasks.

B. Phase 2: Canonical Pattern Clustering

To understand the structural properties of informative strings, we analyzed the *Canonical Patterns* (Pauli strings stripped of identities and sorted, e.g., $X_1 Z_3 \rightarrow XZ$). We trained models across varying dimensions ($N = 3, 4, 5, 6$) on the 20 Newsgroups dataset.

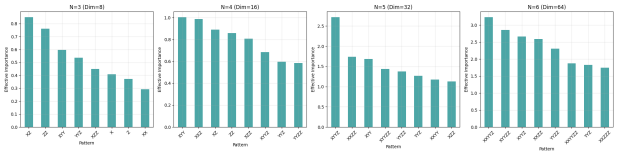

Fig. 1. Canonical Pattern Dominance across Dimensions ($N = 3, 4, 5, 6$). Note how specific patterns (like Z-type) consistently appear as high-importance features, suggesting structural sparsity.

Result: We observed that specific interaction patterns (primarily low-weight Z-type and XZ-type interactions) consistently dominated the feature importance mass, independent of N. This confirmed that the signal is sparse and structured, not random.

C. Phase 3: Toward Data-Adaptive Selection

The lack of a universal static set (Phase 1) and the existence of strong structural signals (Phase 2) led us to the hypothesis that the optimal basis is *data-dependent* but *predictable*. This motivated the development of the Spectral Moment Pauli Generator.

V. Experiments

A. Setup

evaluate on the **20 Newsgroups** dataset (Binary: *alt.atheism* vs *soc.religion.christian*). This high-dimensional text data is ideal for testing scalability.

- **Preprocessing**: TF-IDF, PCA to 2^N dims ($N = 4, 6$), L2 Norm.
- **Model**: ExactSIMClassifier (PyTorch + PennyLane).
- **Ansatz**: StronglyEntanglingLayers ($L = 3$), trained with Adam ($lr = 0.01$).

B. Results: Exact vs Classical Baseline

On the $N = 4$ task, we compared the hybrid Exact SIM model against a Classical SIM baseline (Logistic Regression on the same Pauli expansion).

- **Classical SIM**: 78.48%
- **Exact SIM**: **84.23%**

The hybrid model's superiority validates that the quantum bias term $\langle P \rangle_\theta$ provides a non-trivial inductive bias that aids generalization beyond simple linear combinations.

C. Results: Spectral Pruning (N=4)

We compared selection strategies for $k = 50$ terms (out of 256):

1) **Full Basis (256)**: 83.12%
2) **Spectral Top-50**: **82.93%**
3) **Random Top-50**: 81.45%

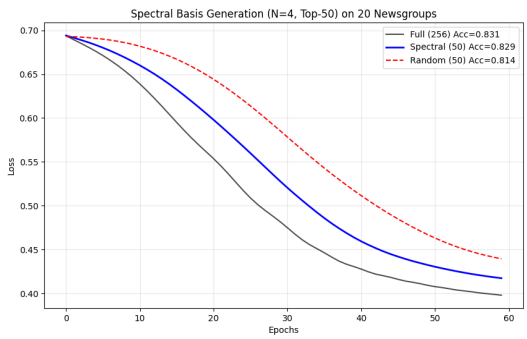

Fig. 2. Training Convergence (N=4). The Spectral Top-50 (Blue) matches the Full Basis (Black) almost perfectly, while Random Top-50 (Red) lags behind.

D. Results: Stress Test and Adaptive Cutoff (N=6)

For $N = 6$ (64 features, 4096 Pauli terms), we performed a compression sweep.

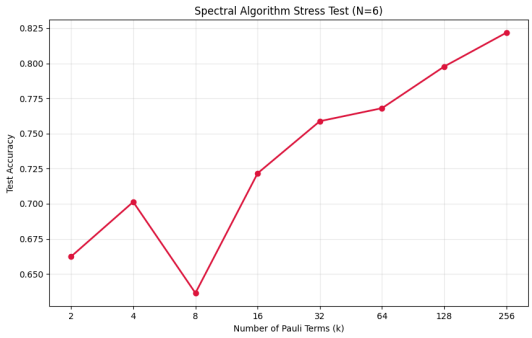

Fig. 3. Accuracy vs Number of Pauli Terms (k) for N=6. The accuracy plateaus around $k = 256$, indicating 94% of terms are redundant.

Using the adaptive cutoff:

- **Full Basis (4096)**: 83.12%
- $\eta = 0.95$ ($k = 1538$): 82.93%
- $\eta = 0.98$ ($k = 1737$): **83.49%**

Remarkably, the $\eta = 0.98$ model slightly outperformed the full model. This suggests that the "tail" of the spectral spectrum contains noise or irrelevant correlations. Pruning them acts as a spectral regularization, preventing the model from overfitting to irrelevant quantum features.

VI. Complexity and Resource Analysis

A key contribution of this work is the reduction of measurement resources required for NISQ implementation.

A. Measurement Complexity

For a general n-qubit Hamiltonian H, estimating the expectation value requires decomposing H into Pauli strings.

- **Naive Flipped Model**: Requires estimating all $4^n - 1$ Pauli terms (full tomography).
- **SIM (Full Basis)**: Requires 4^n measurements. For $N = 10$, this is $\approx 10^6$ terms, which is impractical.
- **Spectral SIM (Proposed)**: Requires estimating only k terms. Our experiments show $k \approx 256$ is sufficient for $N = 6$ (where $4^6 = 4096$).

The complexity reduction is from $\mathcal{O}(4^n)$ to $\mathcal{O}(k)$, where empirically $k \ll 4^n$. This effectively transforms the problem from exponential to sparse polynomial scaling.

B. Classical Overlay

The spectral selection step requires eigendecomposition of a $2^N \times 2^N$ matrix. While this scales as $\mathcal{O}(2^{3N})$, it is a *one-time classical cost* performed during training. The inference cost on the quantum device—which is the expensive resource—is strictly reduced. For very large N, the spectral coefficients can be estimated via Monte Carlo sampling of the classical data, avoiding the full matrix construction.

VII. Conclusion and Future Work

We have introduced **Spectral Pauli Pruning**, a theoretically grounded method for selecting observables in flipped quantum models. By leveraging the spectral decomposition of the covariance difference, we reduced the measurement overhead by $> 90\%$ while maintaining or improving accuracy. Future work will focus on:

1) Scaling this to $N = 10+$ qubits where full Pauli enumeration is impossible, possibly using Monte Carlo estimation of the spectral coefficients.
2) Integrating this with Shadow Tomography for end-to-end efficient training on real hardware.

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